UNNS-U2-FP-SYM

Symbolic Certificates for Bounded Fibonacci Routes
in the Reversible SELECT-II Graph

UNNS Research Collective

Version v001 (supplement to UNNS-U2-FP v001, which is left unchanged)

Abstract

UNNS-U2-FP v001 found, by computer search, bounded routes from F_k to F_{k+1} in the SELECT-II graph for three residue classes of k modulo 12, and left them uncertified. This supplement does three things. (i) It proves those three families exactly, as identities in the quotient ring $\mathbb{Q}[b][a]/(a^2+ab-b^2-\varepsilon)$ generated by the Cassini relation, and gives a short hand proof for the first family from Lucas–Fibonacci identities. (ii) It repeats the route search with a vocabulary of Lucas-factorised labels and finds routes for the *nine remaining classes*, each certified the same way; all 12 residue classes of k modulo 12 are now covered, with certified routes of at most 12 moves, so $d(F_k, F_{k+1}) \le 12$ for every $k \ge 2$. (iii) It tests the requirement raised in the previous review: that every intermediate integer determine its own parameter using only operations allowed by the restricted-selector definition. Under an explicit reading of that definition (Section 6), it does, and the pieces assemble into a finite-control Fibonacci successor selector with 99 path states; an end-to-end simulation that starts from the integer F_k alone and uses no knowledge of k reproduces F_{k+1} for all 670 tested inputs. We do *not* claim the routes are shortest (exact distances for $k \le 29$ are between 2 and 6), we claim no formal-proof-assistant verification, and prime generation is untouched.

1 Scope and status

What is inherited. The graph G, the moves, and the restricted-selector Definition 2.1 (conditions R1–R5) are those of UNNS-U2-FP v001; that package, and U1-BND, U2-001 and U2-SQ, are frozen and are not modified here. Their archive hashes are recorded in the README.

Evidence levels. We separate three kinds of statement.

- *Proved by hand* (Theorem 3): an argument a reader can check line by line.
- *Certified by exact computer algebra*: finite polynomial identities in a commutative ring, plus finite residue checks, carried out with SymPy [1]. This is a computation, not a formal proof; no proof assistant was used.
- *Tested*: numerical replay or simulation on many inputs. Tests are evidence, never a substitute for the first two.

Not claimed. Optimality of any route; that the reading of R3 in Section 6 is the intended one; formal (proof-assistant) verification; anything about primes. No literature search for priority was carried out.

2 Setting

Vertices of G are positive integers. For $N \geq 1$, the move $+N$ sends $s \mapsto s(N+1)^2/N$ and is legal iff $N \mid s$; the move $-N$ sends $s \mapsto sN/(N+1)^2$ and is legal iff $(N+1)^2 \mid s$. They are mutually inverse, so G is undirected. A *route* is a sequence of signed labels (σ_i, N_i). Write d for graph distance.

Lemma 1 (legality from integrality)**.** *Let s, t, N be positive integers. If $t = s(N+1)^2/N$ then $N \mid s$; if $t = sN/(N+1)^2$ then $(N+1)^2 \mid s$. Hence a step is legal as soon as its source, target and label are positive integers satisfying the defining equation.*

Proof. $\gcd(N, (N+1)^2) = 1$. From $Nt = s(N+1)^2$ we get $N \mid s(N+1)^2$, so $N \mid s$. From $(N+1)^2 t = sN$ we get $(N+1)^2 \mid sN$, so $(N+1)^2 \mid s$. □

Definition 2.1 as used here. A restricted selector has a finite control set Q (R1); its labels are computed by bounded straight-line programs over $+, -, \times$, exact division by constants, comparison and fixed-order integer roots (R2); it recovers parameters only by inverting a strictly increasing polynomial or a fixed algebraic closed form (R3); it uses no primality test, factorisation, gcd, divisor enumeration or candidate search (R4); and its labels come from a finite list of certified identities (R5). The reading of R3 adopted in Section 6 is stated explicitly there.

3 Parametrisation and the Cassini ring

For $h \geq 2$ put $a = F_{h-1}$, $b = F_h$, $c = a + b = F_{h+1}$, $L_h = 2a + b$, $L_{h+1} = a + 3b$, and $\varepsilon = (-1)^h$. Cassini's identity reads

$$a^2 + ab - b^2 = \varepsilon. \tag{1}$$

Moreover $F_{2h} = b(2a + b)$, $F_{2h+1} = b^2 + c^2$, $F_{2h+2} = c(a + 3b)$ (these follow from $F_{2h} = F_h L_h$ and $F_{2h+1} = F_h^2 + F_{h+1}^2$, and were also checked numerically for $2 \leq h \leq 119$). So for $k = 2h$ one has $(x, y) = (F_k, F_{k+1}) = (b(2a + b), b^2 + c^2)$ and for $k = 2h + 1$ one has $(x, y) = (b^2 + c^2, c(a + 3b))$.

Let $O_\varepsilon = \mathbb{Q}[b][a]/(a^2 + ab - b^2 - \varepsilon)$. Every element has a unique normal form $p_0(b) + p_1(b)\, a$. The evaluation $b \mapsto F_h$, $a \mapsto F_{h-1}$ is a ring homomorphism $O_\varepsilon \to \mathbb{Q}$ for every h with $(-1)^h = \varepsilon$, by (1). Consequently *an identity between elements of O_ε holds as an identity between integers for every such h.* The fraction field of O_ε is a quadratic extension of $\mathbb{Q}(b)$, since $5b^2 + 4\varepsilon$ is not a square there; this is only used to compute quotients when searching for certificates. The certificates themselves are checked by multiplication in O_ε (verifier item S3b), so no division is trusted.

Proposition 2 (ring certificate)**.** *Let h range over a residue class modulo 6 (so ε is fixed). Suppose $s_0 = x, s_1, \ldots, s_m = y$ and $N_1, \ldots, N_m$ lie in O_ε and satisfy, for each i,*

$$s_i N_i = s_{i-1}(N_i + 1)^2 \quad \textit{(sign +)} \qquad \textit{or} \qquad s_i(N_i + 1)^2 = s_{i-1} N_i \quad \textit{(sign −)}$$

in O_ε. If every s_i and N_i evaluates to a positive integer at every $h \geq h_0$ of the class, then (σ_i, N_i) is a legal route of length m from F_k to F_{k+1} for all such h.

Proof. The identities survive evaluation; Lemma 1 gives legality. □

Checking the hypotheses. *Identity:* finite multiplication in O_ε. *Integrality:* all coefficients have constant denominators D (largest denominator over the class: 4 or 12); the value at h is a fixed polynomial in (F_{h-1}, F_h), so its integrality depends only on (F_{h-1}, F_h) mod D, which is periodic in h with the Pisano period $\pi(D)$ [2]; we evaluate over a full period. *Positivity:* for $h \geq 3$ the ratio a/b

lies in $[\frac{1}{2}, \frac{2}{3}]$, and an element $p_0 + p_1 a$ is linear in a, so positivity for $b \geq 40$ follows from positivity of $p_0 + p_1 r b$ at the two endpoints $r = \frac{1}{2}, \frac{2}{3}$, which we test by checking that all coefficients of the polynomial shifted to $b = 40 + t$ are non-negative. This certifies every class for $b \geq 40$, i.e. $h \geq 10$, i.e. $k \geq 20$; smaller k are covered by the exact table of Section 7.

4 The three families, proved

Theorem 3 (family $k \equiv 4 \pmod{12}$)**.** *For $k = 2h$ with $h \equiv 2 \pmod 6$ and $h \geq 10$, the five moves*

$$+1, \quad +x, \quad +\tfrac{x+3y-2}{4}, \quad -(2y - x + 2), \quad -y$$

form a route from $x = F_k$ to $y = F_{k+1}$.

Proof. Here $\varepsilon = +1$. With $H = L_{h+1} = a + 3b$ and $L = L_h = 2a + b$, direct expansion gives

$$\begin{aligned} x + 1 - aH &= -(a^2 + ab - b^2 - 1), & cL - (y + 1) &= a^2 + ab - b^2 - 1, \\ H^2 - (x + 3y - 2) &= -2(a^2 + ab - b^2 - 1), & L^2 - (2y - x + 2) &= 2(a^2 + ab - b^2 - 1), \\ a(H^2 + 4) - c(L^2 + 1) &= -(3a - b)(a^2 + ab - b^2 - 1). \end{aligned}$$

All five right-hand sides vanish by (1). Hence $x+1 = aH$, $y+1 = cL$, $x+3y-2 = H^2$, $2y-x+2 = L^2$ and $a(H^2 + 4) = c(L^2 + 1)$. Since $h \equiv 2 \pmod 6$, $3 \mid h + 1$ and L_{h+1} is even, so $N := H^2/4$ is an integer. The route is

$$\begin{aligned} x &\xrightarrow{+1} 4x \xrightarrow{+x} 4(x+1)^2 = 4a^2H^2 = 16a^2N \\ &\xrightarrow{+N} 16a^2(N+1)^2 = a^2(H^2+4)^2 = c^2(L^2+1)^2 \\ &\xrightarrow{-L^2} c^2L^2 = (y+1)^2 \xrightarrow{-y} y. \end{aligned}$$

Each move is legal by Lemma 1: every state and label is a positive integer, and the last two sources are multiples of $(L^2 + 1)^2$ and $(y + 1)^2$ respectively. □

The analogous closed forms for $k \equiv 7$ ($k = 2h + 1$, $h \equiv 3 \pmod 6$): $2x + y + 2 = L_{h+1}^2$, $3x - y - 2 = L_h^2$, $x + 1 = F_h L_{h+1}$, $L_h F_{h+2} = y - 1$) and for $k \equiv 10$ ($k = 2h$, $h \equiv 5 \pmod 6$): $x+3y-2 = 5F_{h+1}^2$, $2y-x+2 = 5F_h^2$, $x+1 = F_{h+1}L_{h-1}$, $y+1 = F_h L_{h+1}$) were stated in the note that motivated this supplement; we re-checked them numerically (item S1, 269 cases) and, independently of any closed form, proved the full chains in O_ε (item S2). The routes have 7 moves each: $+x, +(2x+y+2), -\frac{3x-y-2}{4}, -\frac{y-3}{2}, -1, +\frac{y-3}{6}, -\frac{y}{3}$ and $+1, +4, +x, +\frac{x+3y-2}{4}, -(2y-x+2), -4, -y$.

Theorem 4 (three families, certified)**.** *Proposition 2 applies, with all hypotheses verified exactly, to the routes of lengths $5, 7, 7$ for $k \equiv 4, 7, 10 \pmod{12}$ (for $h \geq 10$).* (Computer-algebra certificate; verifier items S2, S3, S3b.)

5 Lucas-factorised search and the nine remaining classes

UNNS-U2-FP v001 searched a vocabulary of linear forms in (x, y) and found routes only for $k \equiv 4, 7, 10$. The factorisations of $F_n \pm 1$ into products of a Fibonacci and a Lucas number suggest enlarging the vocabulary: labels were taken from linear forms $(ua+vb+w)/d$ in (a, b), from products $\lambda\, P\, Q$ with $\lambda \leq 5$ and P, Q drawn from $\{F_{h-1}, F_h, F_{h+1}, F_{h+2}, F_{h+3}, L_{h-1}, L_h, L_{h+1}\}$ (optionally shifted by ± 1), and from the old (x, y) forms; about 1800 values per class. As before, a bidirectional

exact search is run on state tuples evaluated at three sample points of the class, so a hit is a *candidate*, not a theorem. Classes $2, 3, 6, 8$ were found at depth ≤ 8, classes $0, 1, 5, 9$ at depth 8–10. Class 11 was not reached by the full vocabulary within the node caps; it was found only after the vocabulary was reduced to about 1500 values (Section 8).

Every candidate was then (a) replayed on all k of its class below 1500 (122 values per class, all passed), (b) converted to ring elements by exact interpolation in the basis $\{b^j, ab^j\}$ and verified at all sample points, and (c) certified by Proposition 2. Interpolation is only a discovery device: the certificate is the ring identity.

k mod 12	steps	forward	sign pattern	source
0	8	3	+++−+−−−	(a, b) vocabulary
1	10	5	++++−−−+−−	(a, b) vocabulary
2	8	4	++++−−−−	(a, b) vocabulary
3	8	4	++−+−−+−	(a, b) vocabulary
4	5	2	+++−−	(x, y) vocabulary
5	10	5	+++−−+−+−−	(a, b) vocabulary
6	8	4	++−+−+−−	(a, b) vocabulary
7	7	4	++−−−+−	(x, y) vocabulary
8	6	2	++−+−−	(a, b) vocabulary
9	10	5	++−++−+−−−	(a, b) vocabulary
10	7	4	++++−−−	(x, y) vocabulary
11	12	6	++−+−+−++−−−	(a, b) vocabulary

Table 1: Certified routes by residue class of k modulo 12. "Forward" is the number of moves found from the F_k side before the two searches met. The explicit labels are in the Appendix.

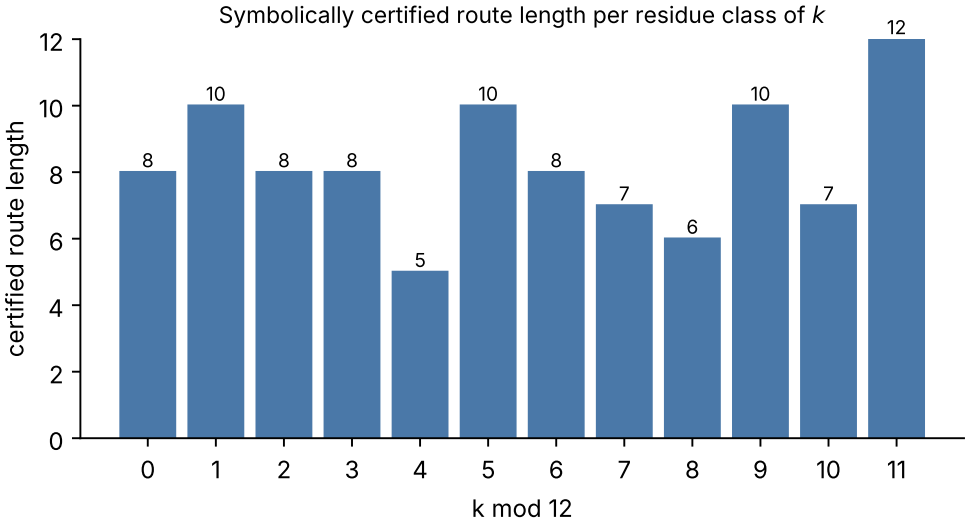


Figure 1: Length of the certified route for each residue class.

Theorem 5 (all twelve classes)**.** *For each residue class of k modulo* 12 *listed in Table 1 there is a route of the stated length from F_k to F_{k+1}, valid for all $k \geq 20$ in the class.* (Computer-algebra certificate, verifier items S3 and S3b: identities in O_ε; integrality over a full Pisano period; positivity for $b \geq 40$; strict monotonicity of all states.) *As an independent test, every individual step of every*

route was checked to be an edge of G with the separate predicate `is_edge` *for all $k < 340$ in the class (2509 steps; item S4).*

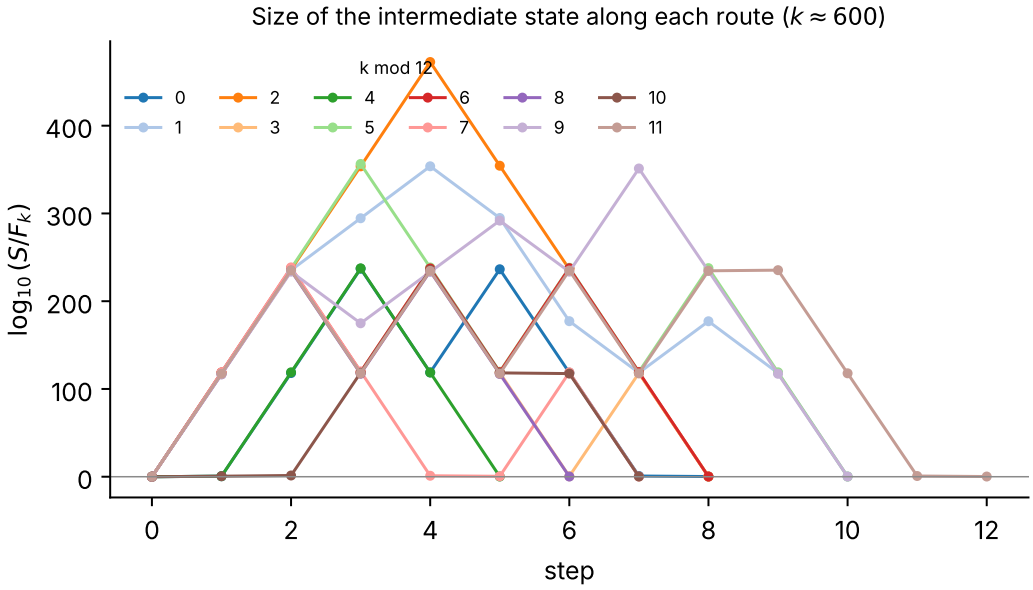


Figure 2: Intermediate states are large: along every route the integer rises to roughly F_k^2–F_k^5 and returns. This is how the routes exploit the squares $(x+1)^2$, $(y+1)^2$.

6 Do the intermediate states determine their parameters?

A bounded route does not by itself give a restricted selector: the label N_i must be computed from the *current integer* and the finite control state, without access to k. We show this is possible for the routes above, under the following explicit reading of R3.

Reading of R3 used. The allowed inversions are (i) fixed-order integer roots and (ii) inversion by bisection of a function $G(t)$ that is a fixed polynomial in t and $A(t)$, where $A(t) = \lfloor(\mathrm{isqrt}(5t^2 + 4\varepsilon) - t)/2\rfloor$, provided G is strictly increasing on its domain. For $t = F_h$ one has $A(t) = F_{h-1}$ exactly. Whether this falls under "a strictly increasing polynomial or a fixed algebraic closed form" is a matter of reading Definition 2.1; if only a one-variable polynomial is admitted, the statements below do not apply.

Dispatch. For $x = F_k$ $(k \geq 2)$, $5x^2 + 4(-1)^k = L_k^2$ and $y = F_{k+1} = (x + L_k)/2$. Hence from x alone one computes L_k (testing which of $5x^2 \pm 4$ is a square) and y. The map $k \mapsto (F_k, F_{k+1})$ mod 24 is injective on a period because the recurrence is invertible, so (x, y) mod 24 determines k mod 24, hence the class k mod 12 and ε.

Proposition 6 (recovery)**.** *Fix a class and a step i of its route, with state $s_i = p_0(b) + p_1(b)a$. Let $G_i(t) = p_0(t) + p_1(t)A(t)$ for integers $t \geq 40$. Then G_i is strictly increasing, $G_i(t) \geq t$, and $G_i(F_h) = s_i$ at the true parameter. Hence F_h is the unique $t \geq 40$ with $G_i(t) = s_i$, and bisection on $[40, s_i]$ finds it.*

Proof. A is non-decreasing and integer-valued, since $\mathrm{isqrt}(5t^2+4\varepsilon)$ grows by 2 or 3 per unit step. For each state one checks (shifted-coefficient test, item S3b and the monotonicity flag of S3) that $p_1 \geq 0$ and $p_1' \geq 0$ for $t \geq 40$ and that $p_0' > 0$ and $p_0(t) \geq t$ there. Then $G_i(t{+}1) - G_i(t) \geq p_0(t{+}1) - p_0(t) > 0$ and $G_i(t) \geq p_0(t) \geq t$. □

Simulation. A selector was implemented that receives only the integer F_k and keeps only the control pair (class, step): it dispatches, then at each step recovers b by bisection, evaluates $N_i = q_0(b) + q_1(b)A(b)$ exactly, checks legality, and applies the move. For all 670 inputs F_k with $30 \leq k < 700$ it ended at F_{k+1} with no failure; the functions G_i were also checked to be strictly increasing on $40 \leq t < 2000$ (item S5). Labels use only $+, -, \times$, division by the constants 2, 3, 4 that occur as coefficient denominators, comparison, and isqrt; no primality test, factorisation, gcd or divisor search occurs. The path part of the control set has 99 states, to which come the start state and the finite table for small k. The labels in the Appendix are the finite certified list required by R5.

Theorem 7 (finite-control Fibonacci successor selector)**.** *Under the reading of R3 above, there is a selector satisfying R1–R5, with control set of size 99 on the path part plus the start state and an explicit table for $k \leq 29$, that on input F_k ($k \geq 2$) outputs a route of at most 12 moves ending at F_{k+1}.*

Proof. For $k \leq 29$ use the table of Section 7. For $k \geq 30$ we have $h \geq 15$, so $b = F_h \geq 40$ and every state's parameter is recoverable by Proposition 6; dispatch gives the class; the labels $N_i(b)$ are the certified ring elements of Theorem 5; legality and the endpoint follow from Proposition 2. □

The simulation above is a corroboration of this theorem, not part of its proof; the proof rests on the exact certificates and on Proposition 6.

7 Small k and the resulting bound

For $2 \leq k \leq 29$ the exact distance $d(F_k, F_{k+1})$ was computed by exhaustive bidirectional search (neighbours enumerated through factorisation), and an explicit path of that length was recorded and re-checked edge by edge with the independent predicate `is_edge` (item S6):

d	2	3	4	5	6
number of $k \in [2, 29]$	1	0	3	17	7

So the certified routes of 5–10 moves are *not* shortest; the true distances are at most 6 for $k \leq 29$, and we make no claim about their value for larger k beyond the upper bound.

Corollary 8. *For every $k \geq 2$, $d(F_k, F_{k+1}) \leq 12$.*

Caution about interpretation. In U2-FP v001 ratio-matched non-Fibonacci pairs had a similar distance distribution, so boundedness of d is not special to Fibonacci numbers. What is special here is *uniformity*: one explicit finite list of integer-valued label formulas, proved for infinitely many k at once, with parameters recoverable from the state.

8 Class 11: how the last route was found

Class $k \equiv 11 \pmod{12}$ resisted the full vocabulary. The attempts are listed in `data/search_record.json`; in short, with about 1800 vocabulary values the bidirectional search reached its node cap before the two frontiers met at depth 10 (modulus 12; and, separately, moduli 24 with $r = 11$ and $r = 23$) and at depth 12 with three million nodes. Reducing the vocabulary to the 1536 values with small coefficients (linear forms with $|u|, |v| \le 2$, $|w| \le 1$, denominators $1, 2, 4$, all products λPQ, the small constants) lowered the branching factor, and the same search met after about 1070 s at total depth 12 (six moves from each side). The candidate replayed correctly on 122 values of k and was then certified like the others. Two lessons: the obstacle was the branching factor, not the expressiveness of the vocabulary; and the route has 12 moves although the exact distance for $k = 11$ and $k = 23$ is 5 and 6, so a considerably shorter route for this class very probably exists.

9 Conclusions and next steps

(1) The three families of UNNS-U2-FP v001 are now proved. (2) The nine remaining classes now have certified routes, so all 12 residue classes of k are covered, with routes of at most 12 moves. (3) Intermediate states determine their parameters under the reading of R3 given above, and the pieces assemble into a finite-control Fibonacci successor selector (Theorem 7). (4) The decisive remaining tasks are: settle whether the reading of R3 in Section 6 is acceptable, or replace the bisection by inversion of a one-variable polynomial; search for shorter routes (the certified lengths 5–12 exceed the exact distances 2–6 found for $k \le 29$); and re-verify the exact certificates in a proof assistant. Prime generation remains a separate open problem.

References

[1] A. Meurer et al., SymPy: symbolic computing in Python, *PeerJ Computer Science* **3**:e103 (2017).

[2] D. D. Wall, Fibonacci series modulo m, *Amer. Math. Monthly* **67** (1960), 525–532.

[3] S. Vajda, *Fibonacci and Lucas Numbers, and the Golden Section*, Ellis Horwood, 1989.

[4] UNNS Research Collective, UNNS-U2-FP v001 (frozen); with U1-BND v001, U2-001 v001, U2-SQ v001.

A Route labels

Labels as found by the search; F_j, L_j with $h = \lfloor k/2 \rfloor$. Moves $1, \ldots, n_f$ are read forward from F_k with S the state before the move; the later moves were found from the F_{k+1} side, and there S is the state *after* the move. The ring-element form of every N_i and s_i is in `data/class_certificates.json`.

Class $k \equiv 0 \pmod{12}$**.**

1	+	$N = 1$
2	+	$N = S/4$
3	+	$N = S/F_{h-1}^2$
4	−	$N = 4F_hF_{h+2}$
5	+	$N = F_{h+1}^2 - 1$
6	−	$N = -x + 2y + 2$
7	−	$N = S/4$
8	−	$N = 1$

Class $k \equiv 1 \pmod{12}$**.**

1	+	$N = S$
2	+	$N = S/(3x - y + 2)$
3	+	$N = 2F_{h-1} + 3F_h$
4	+	$N = 2F_{h-1} + 3F_h + 1$
5	−	$N = \dfrac{2F_{h-1} + 3F_h + 2}{2} - 1$
6	−	$N = F_{h+1}^2$
7	−	$N = F_{h-1} + F_h - 1$
8	+	$N = \dfrac{F_{h-1} + 2F_h + 1}{2} - 1$
9	−	$N = F_{h-1} + 2F_h$
10	−	$N = S$

Class $k \equiv 2 \pmod{12}$**.**

1	+	$N = S$
2	+	$N = F_{h+1}^2$
3	+	$N = \dfrac{-2x + 2y + 2}{2}$
4	+	$N = 4L_{h+1}L_{h-1}$
5	−	$N = 4F_h^2$
6	−	$N = L_{h-1}^2$
7	−	$N = y + 1$
8	−	$N = S$

Class $k \equiv 3 \pmod{12}$**.**

1	+	$N = S$
2	+	$N = 2x + y + 2$
3	−	$N = F_h^2 - 1$
4	+	$N = 4F_{h-1}F_{h+1}$
5	−	$N = 4L_h^2$
6	−	$N = S/4$
7	+	$N = \sqrt{S/(12)} - 1$
8	−	$N = S/3$

Class $k \equiv 4 \pmod{12}$**.**

1	+	$N = 1$
2	+	$N = S/4$
3	+	$N = \dfrac{x+3y+2}{4} - 1$
4	−	$N = -x + 2y + 2$
5	−	$N = S$

Class $k \equiv 5 \pmod{12}$**.**

1	+	$N = S$
2	+	$N = 3x - y + 2$
3	+	$N = 2x + y + 2$
4	−	$N = F_{h-1}F_{h+1} - 1$
5	−	$N = y - 2$
6	+	$N = \dfrac{y}{2} - 1$
7	−	$N = \dfrac{-x + 2y - 1}{2}$
8	+	$N = F_{h-1}F_{h+2} - 1$
9	−	$N = 3x - y + 2$
10	−	$N = S$

Class $k \equiv 6 \pmod{12}$**.**

1	+	$N = S$
2	+	$N = F_{h+1}^2$
3	−	$N = \dfrac{-2x + 2y + 2}{4} - 1$
4	+	$N = \dfrac{-2x + 2y - 2}{8}$
5	−	$N = \dfrac{-2x + 2y + 2}{4}$
6	+	$N = -x + y + 1$
7	−	$N = S/\left(x + 3y + 2\right)$
8	−	$N = S$

Class $k \equiv 7 \pmod{12}$**.**

1	+	$N = S$
2	+	$N = 2x + y + 2$
3	−	$N = \dfrac{3x - y + 2}{4} - 1$
4	−	$N = \dfrac{y - 1}{2} - 1$
5	−	$N = 1$
6	+	$N = \sqrt{S/\left(12\right)} - 1$
7	−	$N = S/3$

Class $k \equiv 8 \pmod{12}$**.**

1	+	$N = S$
2	+	$N = 4L_{h+1}^2$
3	−	$N = 4F_hF_{h+2}$
4	+	$N = F_{h+1}^2 - 1$
5	−	$N = -x + 2y + 2$
6	−	$N = S$

Class $k \equiv 9 \pmod{12}$**.**

1	+	$N = S$
2	+	$N = S/(3x - y + 2)$
3	−	$N = 2F_{h-1} + 3F_h - 1$
4	+	$N = \dfrac{2F_{h-1} + 3F_h - 1}{2}$
5	+	$N = \dfrac{2F_{h-1} + 3F_h + 1}{2}$
6	−	$N = \dfrac{2F_{h-1} + 3F_h - 1}{4}$
7	+	$N = F_{h+3}^2 - 1$
8	−	$N = F_{h+1}^2$
9	−	$N = S/(3x - y + 2)$
10	−	$N = S$

Class $k \equiv 10 \pmod{12}$**.**

1	+	$N = 1$
2	+	$N = 4$
3	+	$N = x$
4	+	$N = \dfrac{x + 3y - 2}{4}$
5	−	$N = -x + 2y + 2$
6	−	$N = 4$
7	−	$N = S$

Class $k \equiv 11 \pmod{12}$**.**

1	+	$N = S$
2	+	$N = S/F_h^2$
3	−	$N = \sqrt{S/\left(F_{h+2}^2\right)} - 1$
4	+	$N = S/\left(4F_{h+2}^2\right)$
5	−	$N = \sqrt{S/\left(F_{h+2}^2\right)} - 1$
6	+	$N = F_{h+2}^2$
7	−	$N = S/(2L_{h+1}L_{h-1})$
8	+	$N = \sqrt{S/(4L_{h-1}F_{h+3})} - 1$
9	+	$N = 3$
10	−	$N = L_{h-1}F_{h+3}$
11	−	$N = S/12$
12	−	$N = 1$